# Happy Hallowe'en - It's National More SSH Day!

*The apostrophe honors those who have tricked and treated before us.*

## Setup changes for Proxmox

I think you'll like having SPICE display drivers. To review, here's how to do it:

1. Shut down your VM
2. Change the CPU type to `x86-64-v2-AES`
3. Ensure that your display is set to `VirtIO-GPU` (with 256 MB of memory)
4. Download the `virtviewer` software for *your* computer (not your VM) and install it
5. Profit

> NOTE: After you are using SPICE, you will want to change the scaling factor for Ubuntu desktop to 200%. I can show you how to do this.

## Installing and enabling SSH Server on your VM

SSH Server can run on almost any operating system. To set it up on your Ubuntu Desktop VM, do the following:

```
# what does the next command do?
sudo apt update
sudo apt install openssh-server
```

To start SSH Server and make it run at every Ubuntu startup:

```
# start ssh server:
sudo systemctl start ssh

# configure ssh server to auto-start when ubuntu starts:
sudo systemctl enable ssh
```

If you prefer, you can do both of these things with one command instead of two:

```
# start ssh server and configure it to auto-start with a single command:
sudo systemctl enable --now ssh
```

## Adding SSH users to your Ubuntu VM

Next, we want to add users that can login to your VM using SSH.

1. In the GUI, go to **Settings** → **System** → **Users** → **Unlock** → **Add User**
2. Fill in the **Name** field and the **Username** field will autofill
3. Do **not** make the user an Administrator

4. Click Set password now and give the user an easy password. You will probably give this to another student later

## Protecting root

This is a tricky thing to do. For example, if an account that can use `sudo` to become `root` is compromised, an attacker could install an altered version of `sudo` that would log keystrokes to obtain the user's password, and with it, the ability to `sudo` to root.

## Homework

> NOTE: never use AI to solve any of the homework problems UNLESS:
>
> - you have tried yourself and cannot think of what to try next; and
> - you ask the AI to explain it to you.
>
> If you ask the AI for help, tell it:
>
> - your school level ("middle school," or "high school," or your age);
> - how much Linux experience you have; and
> - exactly what you don't understand.
>
> Then ask it to explain its answer step by step. If a step doesn't make sense, say so and ask it to explain that step differently.
>
> If you're still confused after that, you can post your questions on Zulip, or bring them to class.

### Homework #1: finish SSH setup

The most important thing is to finish setting up an SSH server on your VM. We need this so that you can work back and forth with a "lab partner" and so we'll be at the same setup level.

- Running `systemctl status ssh` shows that SSH is running and enabled.
- You have created a user on your VM.
- You've tested that SSH works for your new user. For example, if your user is named `my_sshuser`, you can check their SSH access by running `ssh my_sshuser@<your_IP_address>`, supplying the IP of your own VM.

### Homework #2: SSH settings

As we saw today, settings for the SSH server software are set in `/etc/ssh/sshd_config/` and `/etc/ssh/sshd_config.d/`. Find a resource online that explains what the most important settins are and what to set them to. Write them down so we can discuss them next week.

### **Extra Credit:** Locking down an SSH user

You can restrict a user so that:

- they can log in to your computer using SSH; but
- they cannot log in to your computer using the Desktop login; and
- they cannot login in to your computer using the Console.

Figure out how, then write up what you did. If you know how to make a GitHub PR, submit your write-up to the cdc-2027 repo under the `docs\ssh-lockdown` folder.